



On the dynamics of viscoelastic discontinuous beams



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ABSTRACT

This paper concerns the dynamics of beams with an arbitrary number of Kelvin–Voigt viscoelastic rotational joints, translational supports, and attached lumped masses. Using the theory of generalized functions to treat the discontinuities of the response variables, the free vibration problem is solved upon deriving exact closed-form eigenfunctions, that inherently fulfill the required conditions at the discontinuity points. The forced vibration response is computed in time and frequency domain by modal superposition, based on appropriate orthogonality conditions of the eigenfunctions.

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1. Introduction

Engineering applications frequently involve beams with joints, supports and attachments. Within a standard 1D formulation of the governing equations, beams of this sort are commonly referred to as *discontinuous* beams, because pertinent discontinuities have to be considered in the response variables. Especially in structural dynamics, rotational joints (RJs), translational supports (TSs) and attached lumped mass (LMs) play an important role and, in many cases, both flexibility and damping shall be accounted for in RJs and TSs. In this context, a typical constitutive law of RJs and TSs involves linear elasticity and viscous damping, corresponding to a Kelvin–Voigt (KV) viscoelastic law. Examples are in TSs modeling external dampers for vibration control (Shukla and Datta, 1999; FEMA 273), RJs modeling dissipating beam-to-column connections (Xu and Zhang, 2001), RJs modeling bolted or welded joints with imperfections or damage (Kawashima and Fujimoto, 1984; Sekulovic et al., 2002).

A classical solution to the eigenvalue problem of beams with KV viscoelastic RJs, TSs and LMs involves expressing, over uniform segments between consecutive discontinuity points, the vibration response in a trigonometric form with four unknown constants, and enforcing matching conditions at the discontinuity points along with the boundary conditions (B.C.). In this context, a complex modal analysis is generally required, being the damping non-proportional (Veletsos and Ventura, 1986). A similar approach should be pursued to compute the frequency response function (FRF). However, the equations rapidly increase with the number of discontinuities, and must be updated whenever their locations change along the beam axis. These are significant drawbacks for design, optimization and sensitivity analyses, which require building different solutions for different potential system parameters. Motivated by these issues, alternative solutions have been sought.

Approximate solutions have been obtained by an assumed-mode method based on eigenfunctions of the *bare* beam, in conjunction with a variational formulation. In this context, the characteristic equation (CE) has been built as determinant of an appropriate matrix by Cha (2005) for a EB beam with multiple KV viscoelastic TSs, and has been derived in a closed form by Gürgöze (1998) for a EB cantilever beam carrying a tip mass and supported by an in-span damper. Also, Gürgöze and Erol (2002) have built an approximate time domain response function for an EB cantilever beam carrying a tip mass, with an in-span damper and a fixed support.

In some cases exact solutions have been also found. For instance, an exact closed-form CE has been derived by Chang et al. (2001) based on the Laplace transform, for a EB carrying an arbitrary number of elastic or viscous TSs and LMs. Following a transfer matrix approach, Sorrentino et al. (2003) have built the CE as determinant of a 4×4 matrix, regardless of the number of discontinuity points, the FRF and the impulse response function (IRF) by modal superposition upon demonstrating orthogonality conditions of the eigenfunctions, for KV

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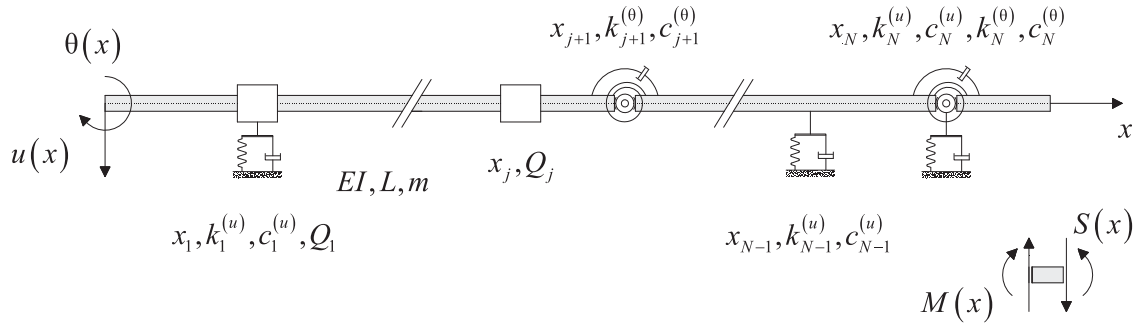


Fig. 1. Beam with arbitrary number of RJs, TSs and LMs.

viscoelastic TSs and LMs in either EB or Timoshenko (TM) beams (Sorrentino et al., 2003, 2007). Following a dynamic stiffness matrix approach, Hong and Kim (1999) have built the CE as determinant of a beam dynamic stiffness matrix obtained by assembling those of the uniform segments between two consecutive discontinuity points, and a closed-form FRF matrix and a IRF matrix by modal superposition, for TM beams with KV viscoelastic RJs, TSs and LMs (Hong and Kim, 1999). However, by this approach the size of the matrices inevitably increases with the number of discontinuity points.

With the aim of contributing to the research effort in this field, this paper will focus on EB beams with an arbitrary number of KV viscoelastic RJs, TSs and LMs. Using the theory of generalized functions to treat the discontinuities of the response variables (Caddemi and Caliò, 2009, 2013; Caddemi et al., 2013a; Failla and Santini, 2007, 2008; Failla, 2011; Failla and Impollonia, 2012), a solution will be found to free and forced vibrations. In particular, the exact CE will be obtained as determinant of a 4×4 matrix, upon deriving exact closed-form expressions for the eigenfunctions, that inherently fulfill the required conditions at the discontinuity points. Hence, orthogonality conditions of the eigenfunctions, closed-form modal impulse response functions (MIRFs), and closed-form modal frequency response functions (MFRFs) will be obtained generalizing a theoretical framework previously devised by Oliveto et al. (1997) for a EB beam with viscous RJs at the ends. The MIRFs and MFRFs will be used to solve forced vibrations problems, using the complex modal superposition principle as presented by Oliveto et al. (1997).

2. Free vibration equations by theory of generalized functions

Consider the EB beam in Fig. 1. Be x the longitudinal axis (positive rightward), L the length, EI the flexural stiffness, m the mass per unit length. Denote by $u(x)$ and $\theta(x)$ deflection and rotation of the cross section, by $M(x)$ and $S(x)$ bending moment and shear force. Positive signs are shown in Fig. 1.

It is assumed that an arbitrary number of KV viscoelastic RJs, TSs, LMs are arbitrarily located at abscissas x_j 's along the beam axis, for $j = 1, 2, \dots, N$, $0 < x_1 < \dots < x_j < \dots < x_N < L$; x_j 's will be referred to as discontinuity points. Stiffness and damping parameters are $k_j^{(\theta)}$ and $c_j^{(\theta)}$ for the j th RJ, $k_j^{(u)}$ and $c_j^{(u)}$ for the j th TS; Q_j is the j th LM. Equations will be written for the most general case when a RJ, a TS and a LM occur simultaneously at every discontinuity point. At any rate, straightforward changes can be made, whereas any of the latter is not present at a discontinuity point. Following the approach by Wang and Qiao (2007), who have studied EB beams with discontinuities due to linearly-elastic RJs, TSs, LMs and other type of discontinuities, the free vibration equilibrium equations read

$$\frac{\partial S}{\partial x} + \sum_{j=1}^N V_j \delta(x - x_j) = m \ddot{u}(x); \quad (1)$$

$$\frac{\partial M}{\partial x} = S(x) \quad (2)$$

where dot means derivative with respect to time and bar means generalized derivative; $\delta(x - x_j)$ is the Dirac's delta, given as the generalized derivative of the unit-step function $H(x - x_j)$ (Falsone, 2002)

$$\delta(x - x_j) = \frac{\partial H(x - x_j)}{\partial x}; \quad (3)$$

$$H(x - x_j) = 0, \quad x < x_j; \quad H(x - x_j) = 1, \quad x > x_j. \quad (4a,b)$$

Also, in Eq. (1) V_j is the shear-force discontinuity due to the TS and the LM at x_j , i.e.

$$V_j = V_j^{(u)} + V_j^{(Q)}, \quad (5)$$

$$\text{for } V_j^{(u)} = -k_j^{(u)} u(x_j) - c_j^{(u)} \dot{u}(x_j); \quad V_j^{(Q)} = -Q_j \ddot{u}(x_j), \quad (6a,b)$$

being $V_j^{(u)}$ and $V_j^{(Q)}$ both positive downwards. For $u(x)$ and $\theta(x)$, it can be written that

$$\frac{\partial \theta}{\partial x} = -\frac{M(x)}{EI} + \sum_{j=1}^N r_j \delta(x - x_j); \quad (7)$$

$$\frac{\bar{\partial}u}{\partial x} = \theta(x), \quad (8)$$

where $r_j = \theta(x_j^+) - \theta(x_j^-)$ is the rotation discontinuity at $x = x_j$, such that:

$$M(x_j) = -k_j^{(\theta)} r_j - c_j^{(\theta)} \dot{r}_j \quad (9)$$

The generalized derivatives are introduced in Eqs. (1)–(2) and Eqs. (7)–(8) since the corresponding variables are not differentiable in a classical sense, as a result of the shear-force and rotation discontinuities (Falsone, 2002). Time dependence is omitted for brevity. Eq. (1) through Eq. (9) yield

$$EI \frac{\bar{\partial}^4 u}{\partial x^4} + \sum_{j=1}^N f_j(x) + m\ddot{u}(x) = 0, \quad (10)$$

$$\text{for } f_j(x) = -V_j \delta(x - x_j) - EI \cdot r_j \delta^{(2)}(x - x_j) \quad (11)$$

being $\delta^{(n)}(x - x_j)$ the n th generalized derivative of the Dirac's delta. Eqs. (10)–(11) correspond to Eq. (11) by Wang and Qiao (2007), where additional terms are included for other discontinuities.

3. Eigensolution

The eigenvalue problem is formulated in dimensionless form based on Eq. (10), following a standard separate variable approach. That is, $u(x) = U(x)e^{i\sigma t}$ is set (i = imaginary unit) in Eq. (10) to derive

$$\frac{\bar{d}^4 \psi}{d\xi^4} + \sum_{j=1}^N \phi_j(\omega) - \omega^2 \psi(\xi) = 0, \quad (12)$$

$$\text{for } \phi_j(\omega) = -v_j(\omega)\delta(\xi - \xi_j) - \rho_j(\omega)\delta^{(2)}(\xi - \xi_j) \quad (13)$$

where $\xi = x/L$, $\psi(\xi) = U(x)/L$, $\omega^2 = \sigma^2 m L^4 / EI$. In Eq. (13) $v_j = V_j L^2 / EI$ is the shear-force discontinuity at ξ_j

$$v_j(\omega) = v_j^{(u)}(\omega) + v_j^{(Q)}(\omega), \quad \text{for} \quad (14)$$

$$\begin{aligned} v_j^{(u)}(\omega) &= -\kappa_{eq,j}^{(u)}(\omega) \psi(\xi_j) \quad \text{for } \kappa_{eq,j}^{(u)}(\omega) = \kappa_j^{(u)} + i\omega \gamma_j^{(u)}, \quad j = 1, 2, \dots, N \\ v_j^{(Q)}(\omega) &= \Pi_j \omega^2 \psi(\xi_j), \quad j = 1, 2, \dots, N. \end{aligned} \quad (15a,b)$$

being $\kappa_j^{(u)} = k_j^{(u)} L^3 / EI$, $\gamma_j^{(u)} = c_j^{(u)} L / (mEI)^{1/2}$ and $\Pi_j = Q_j / mL$. In Eq. (13) ρ_j is the rotation discontinuity at ξ_j

$$\rho_j(\omega) = \theta(\xi_j^+) - \theta(\xi_j^-) = -(\kappa_j^{(\theta)} + i\omega \gamma_j^{(\theta)})^{-1} \mu(\xi_j) = -(\kappa_{eq,j}^{(\theta)}(\omega))^{-1} \mu(\xi_j), \quad (16)$$

where $\mu(\xi) = M(x)L/EI$ is the dimensionless bending moment, $\kappa_j^{(\theta)} = k_j^{(\theta)} L / EI$ and $\gamma_j^{(\theta)} = c_j^{(\theta)} L / (mEI)^{1/2}$. It is observed that $\kappa_{eq,j}^{(u)}(\omega)$ and $\kappa_{eq,j}^{(\theta)}(\omega)$ are the dimensionless frequency-dependent stiffness parameter (Cha, 2005) of the j th RJ and TS. Eqs. (12)–(13) correspond to Eq. (13) by Wang and Qiao (2007).

Next, it is seen that the exact solution to Eq. (12) can be found in the general form

$$\psi(\xi) = \psi_{om}(\xi) + \psi_v(\xi) + \psi_\rho(\xi), \quad (17)$$

where $\psi_{om}(\xi)$ is the solution to the homogeneous equation

$$\psi_{om}(\xi) = a_1 e^{-\sqrt{\omega}\xi} + a_2 e^{-i\sqrt{\omega}\xi} + a_3 e^{\sqrt{\omega}\xi} + a_4 e^{i\sqrt{\omega}\xi}, \quad (18a)$$

being a_k unknown integration constants; $\psi_v(\xi)$ and $\psi_\rho(\xi)$ are the particular integrals related to the shear-force discontinuities (ψ_v) and rotation discontinuities (ψ_ρ), given respectively as

$$\psi_v(\xi) = \sum_{j=1}^N \frac{1}{4\omega^{3/2}} [ie^{i\sqrt{\omega}(\xi-\xi_j)} - e^{-\sqrt{\omega}(\xi-\xi_j)} + e^{\sqrt{\omega}(\xi-\xi_j)} - ie^{-i\sqrt{\omega}(\xi-\xi_j)}] v_j H(\xi - \xi_j) \quad (18b)$$

$$\psi_\rho(\xi) = \sum_{j=1}^N \frac{1}{4\sqrt{\omega}} [-ie^{i\sqrt{\omega}(\xi-\xi_j)} - e^{-\sqrt{\omega}(\xi-\xi_j)} + e^{\sqrt{\omega}(\xi-\xi_j)} + ie^{-i\sqrt{\omega}(\xi-\xi_j)}] \rho_j H(\xi - \xi_j) \quad (18c)$$

Eq. (17) is the exact deflection eigenfunction corresponding to the eigenvalue ω . Based on Eq. (17), the other response eigenfunctions can readily be derived by simple differentiation rules of generalized calculus. The rotation eigenfunction $\theta(\xi)$ is obtained as

$$\theta(\xi) = \frac{\bar{d}\psi}{d\xi} = \theta_{om}(\xi) + \theta_v(\xi) + \theta_\rho(\xi) \quad (19)$$

where $\theta_{om}(\xi)$ is the classical first derivative of $\psi_{om}(\xi)$ given by Eq. (18a) (not reported for brevity) and

$$\theta_v(\xi) = -\sum_{j=1}^N \frac{e^{-i\sqrt{\omega}(\xi+\xi_j)}}{4\omega} \cdot (e^{i\sqrt{2\omega}\xi} - e^{i\sqrt{2\omega}\xi_j}) \cdot (e^{\sqrt{2\omega}\xi} - e^{\sqrt{2\omega}\xi_j}) v_j H(\xi - \xi_j) \quad (20a)$$

$$\theta_\rho(\xi) = \sum_{j=1}^N \frac{e^{-i\sqrt{\omega}(\xi+\xi_j)}}{4} \cdot (e^{i\sqrt{2\omega}\xi} + e^{i\sqrt{2\omega}\xi_j}) \cdot (e^{\sqrt{2\omega}\xi} + e^{\sqrt{2\omega}\xi_j}) \rho_j H(\xi - \xi_j) \quad (20b)$$

Notice that, as expected, the particular solution θ_ρ related to the RJs is discontinuous at ξ_j . Taking into account Eqs. (7) and (8), the bending moment eigenfunction $\mu(\xi)$ is derived as

$$\mu(\xi) = -\frac{\bar{d}^2 \psi}{d\xi^2} + \sum_{j=1}^N \rho_j \delta(\xi - \xi_j) = \mu_{om}(\xi) + \mu_v(\xi) + \mu_\rho(\xi) \quad (21)$$

where $\mu_{om}(\xi)$ is the classical second derivative of $\psi_{om}(\xi)$ given by Eq. (18a) and

$$\mu_v(\xi) = -\sum_{j=1}^N \frac{1}{4\sqrt{\omega}} (-ie^{i\sqrt{\omega}(\xi-\xi_j)} + e^{\sqrt{\omega}(\xi-\xi_j)} + ie^{-i\sqrt{\omega}(\xi-\xi_j)} - e^{-\sqrt{\omega}(\xi-\xi_j)}) v_j H(\xi - \xi_j) \quad (22a)$$

$$\mu_\rho(\xi) = -\sum_{j=1}^N \frac{\sqrt{\omega}}{4} (ie^{i\sqrt{\omega}(\xi-\xi_j)} + e^{\sqrt{\omega}(\xi-\xi_j)} - ie^{-i\sqrt{\omega}(\xi-\xi_j)} - e^{-\sqrt{\omega}(\xi-\xi_j)}) \rho_j H(\xi - \xi_j) \quad (22b)$$

Using Eq. (2) and Eq. (21), the shear-force eigenfunction, $\chi(\xi) = S(x)L^2/EI$, can be computed as

$$\chi(\xi) = -\frac{\bar{d}^3 \psi}{d\xi^3} + \sum_{j=1}^N \rho_j \delta^{(1)}(\xi - \xi_j) = \chi_{om}(\xi) + \chi_v(\xi) + \chi_\rho(\xi) \quad (23)$$

where $\chi_{om}(\xi)$ is the classical third derivative of $\psi_{om}(\xi)$ given by Eq. (18a) and

$$\chi_v(\xi) = -\frac{1}{4} e^{-i\sqrt{\omega}(\xi+\xi_j)} (e^{i\sqrt{2\omega}\xi} + e^{i\sqrt{2\omega}\xi_j}) \cdot (e^{\sqrt{2\omega}\xi} + e^{\sqrt{2\omega}\xi_j}) v_j H(\xi - \xi_j) \quad (24a)$$

$$\chi_\rho(\xi) = \frac{\omega}{4} e^{-i\sqrt{\omega}(\xi+\xi_j)} (e^{i\sqrt{2\omega}\xi} - e^{i\sqrt{2\omega}\xi_j}) \cdot (e^{\sqrt{2\omega}\xi} - e^{\sqrt{2\omega}\xi_j}) \rho_j H(\xi - \xi_j) \quad (24b)$$

As expected, the particular solution $\chi_v(\xi)$ related to the TSs and LMs is discontinuous at ξ_j .

At this stage, it is important to point out that the eigenfunctions ψ , θ , μ , χ involve $2N$ unknowns at the discontinuity points ξ_j 's, i.e. the shear-force discontinuities v_j 's and the rotation discontinuities ρ_j 's. Using Eqs. (14)–(16), they can be computed by setting the following identities at ξ_j 's:

$$v_j = -\kappa_{eq,j}^{(u)}(\omega) \psi(\xi_j) + \Pi_j \omega^2 \psi(\xi_j); \quad \rho_j = -(\kappa_{eq,j}^{(\theta)}(\omega))^{-1} \mu(\xi_j); \quad j = 1, 2, \dots, N \quad (25a,b)$$

After some manipulations, Eqs. (25) can be recast in the following form:

$$\sum_{k=1}^{j-1} (D_{jk}^{(\rho\rho)} \rho_k + D_{jk}^{(\rho v)} v_k) + \rho_j = f_\rho(\omega, \xi_j, \mathbf{a}); \quad \sum_{k=1}^{j-1} (D_{jk}^{(v\rho)} \rho_k + D_{jk}^{(vv)} v_k) + v_j = f_v(\omega, \xi_j, \mathbf{a}) \quad (26a,b)$$

where coefficients $D_{jk}^{(\rho\rho)}$, $D_{jk}^{(\rho v)}$, $D_{jk}^{(v\rho)}$, $D_{jk}^{(vv)}$ are closed-form functions of the discontinuity parameters:

$$D_{jk}^{(\rho\rho)} = \frac{\sqrt{\omega}(ie^{i\sqrt{\omega}(\xi_k-\xi_j)} + e^{\sqrt{\omega}(\xi_k-\xi_j)} - ie^{i\sqrt{\omega}(\xi_j-\xi_k)} - e^{\sqrt{\omega}(\xi_j-\xi_k)})}{4\kappa_{eq,j}^{(\theta)}(\omega)} \quad (27a)$$

$$D_{jk}^{(\rho v)} = \frac{-ie^{i\sqrt{\omega}(\xi_k-\xi_j)} + e^{\sqrt{\omega}(\xi_k-\xi_j)} + ie^{i\sqrt{\omega}(\xi_j-\xi_k)} - e^{\sqrt{\omega}(\xi_j-\xi_k)}}{4\kappa_{eq,j}^{(\theta)}(\omega)\sqrt{\omega}} \quad (27b)$$

$$D_{jk}^{(v\rho)} = \frac{(\kappa_{eq,j}^{(u)}(\omega) - \Pi_j \omega^2) (ie^{i\sqrt{\omega}(\xi_k-\xi_j)} + e^{\sqrt{\omega}(\xi_k-\xi_j)} - e^{\sqrt{\omega}(\xi_j-\xi_k)} - ie^{i\sqrt{\omega}(\xi_j-\xi_k)})}{4\sqrt{\omega}} \quad (27c)$$

$$D_{jk}^{(vv)} = \frac{(\kappa_{eq,j}^{(u)}(\omega) - \Pi_j \omega^2) (-ie^{i\sqrt{\omega}(\xi_k-\xi_j)} + e^{\sqrt{\omega}(\xi_k-\xi_j)} - e^{\sqrt{\omega}(\xi_j-\xi_k)} + ie^{i\sqrt{\omega}(\xi_j-\xi_k)})}{4\omega^{3/2}} \quad (27d)$$

Likewise, $f_\rho(\omega, \xi_j, \mathbf{a})$ and $f_v(\omega, \xi_j, \mathbf{a})$ are the following closed-form functions:

$$f_\rho(\omega, \xi_j, \mathbf{a}) = \frac{\omega(e^{-\sqrt{\omega}\xi_j} a_1 - e^{-i\sqrt{\omega}\xi_j} a_2 + e^{\sqrt{\omega}\xi_j} a_3 - e^{i\sqrt{\omega}\xi_j} a_4)}{\kappa_{eq,j}^{(\theta)}(\omega)} \quad (28a)$$

$$f_v(\omega, \xi_j, \mathbf{a}) = \left(-\kappa_{eq,j}^{(u)}(\omega) + \prod_j \omega^2 \right) (e^{-\sqrt{\omega}\xi_j} a_1 + e^{-i\sqrt{\omega}\xi_j} a_2 + e^{\sqrt{\omega}\xi_j} a_3 + e^{i\sqrt{\omega}\xi_j} a_4) \quad (28b)$$

where $\mathbf{a}$ is the vector of a_k 's. The solution to Eqs. (26) is straightforward, since the equations at $\xi = \xi_j$ involve only the unknowns v_k 's and ρ_k 's for $k \leq j$. Therefore, upon computing v_1 and ρ_1 at ξ_1 as

$$\rho_1 = f_\rho(\omega, \xi_1, \mathbf{a}); \quad v_1 = f_v(\omega, \xi_1, \mathbf{a}), \quad (29a,b)$$

v_j and ρ_j can readily be computed as

$$\rho_j = f_\rho(\omega, \xi_j, \mathbf{a}) - \sum_{k=1}^{j-1} (D_{jk}^{(\rho\rho)} \rho_k + D_{jk}^{(\rho v)} v_k), \quad v_j = f_v(\omega, \xi_j, \mathbf{a}) - \sum_{k=1}^{j-1} (D_{jk}^{(v\rho)} \rho_k + D_{jk}^{(vv)} v_k) \quad (30a,b)$$

In this manner, v_j 's and ρ_j 's are all derived as closed-form, linear functions of the integration constants a_k 's. This solution procedure is applicable also when RJs, TSs and LMs do not occur simultaneously at a discontinuity point ξ_j . In this case, $\rho_j = 0$ shall be set at ξ_j where no rotation discontinuity occurs and, similarly, $v_j = 0$ at ξ_j where no shear-force discontinuity occurs. If the shear-force discontinuity is due to a TS or a LM only, $\Pi_j = 0$ or $\kappa_{eq,j}^{(u)} = 0$ shall be set, respectively.

3.1. Characteristic equation

Upon solving Eqs. (26) for v_j 's and ρ_j 's, the exact eigenfunctions ψ , θ , μ and χ are available in a closed form, linearly depending on the integration constants a_k 's. Then, the eigenvalues ω can be computed by enforcing the B.C. and seeking for nontrivial values of a_k 's. This corresponds to set

$$\mathbf{B}\mathbf{a} = \mathbf{0}; \quad (31)$$

$$\det(\mathbf{B}) = 0 \quad (32)$$

where $\mathbf{B}$ is a 4×4 matrix depending on the B.C. Eq. (32) represents the exact CE. It can be implemented in any symbolic package and solved by solution algorithms for nonlinear equations (Sorrentino et al., 2003). Since damping is not proportional, complex eigenvalues ω_k are expected, with the general form

$$\omega_k = \pm p_k + iq_k \quad (33)$$

By introducing (Oliveto et al., 1997)

$$|\omega_k| = \sqrt{p_k^2 + q_k^2}; \quad \zeta_k = \frac{q_k}{\sqrt{p_k^2 + q_k^2}}, \quad (34a,b)$$

Eq. (33) can be rewritten as

$$\omega_k = \pm p_k + iq_k = \pm |\omega_k| \sqrt{1 - \zeta_k^2} + i\zeta_k |\omega_k| \quad (35)$$

As noticed by Veletsos and Ventura (1986), Eqs. (35) are the same as those governing the characteristics roots of a viscously-damped SDOF system with an undamped natural frequency $|\omega_k|$, Eq. (34a), a damping ratio ζ_k , Eq. (34b), and a damped natural frequency $p_k = |\omega_k| \sqrt{1 - \zeta_k^2}$. For this, ζ_k and p_k are generally referred to as k th modal damping ratio and k th damped frequency; $|\omega_k|$ is referred to as the k th pseudo-undamped natural frequency, due to its dependence on the damping parameter $q_k = \zeta_k |\omega_k|$ (Veletsos and Ventura, 1986). It is observed that, for damping levels generally encountered in structural applications, $q_k \geq 0$ and $p_k > 0$ (Veletsos and Ventura, 1986; Oliveto et al., 1997). If $q_k > 0$ and $p_k > 0$, a pair of complex-conjugate eigenfunctions ($\psi_k(\xi), \psi_k^*(\xi)$) correspond to $\omega_k = \pm p_k + iq_k$, Eq. (33) (* = complex conjugate) while, if $q_k = 0$, the mode is real. The physical meaning of real part and imaginary part of a pair of complex-conjugate eigenfunctions ($\psi_k(\xi), \psi_k^*(\xi)$) has been illustrated by Veletsos and Ventura (1986) and details are not repeated for brevity. Here, it is reminded only that the k th modal solution associated with the k th pair of complex-conjugate eigenfunctions ($\psi_k(\xi), \psi_k^*(\xi)$) is represented by the superposition of two harmonic motions shifted by 90° , and scaled by an exponentially decaying function (see Eq. (21) in the paper by Veletsos and Ventura (1986)). The circular frequency of the component harmonic motions is the damped natural frequency $p_k = |\omega_k| \sqrt{1 - \zeta_k^2}$, while the exponential function decays with $q_k = \zeta_k |\omega_k|$. The beam configurations associated with the two component harmonic motions are given by the real part and imaginary part of the complex-conjugate eigenfunctions ($\psi_k(\xi), \psi_k^*(\xi)$). The differences with respect to the classical modes of undamped systems are (i) the beam configuration repeats itself at intervals $T_k = 2\pi/p_k$, but scaled by the exponentially decaying function; (ii) during the period T_k , the beam configuration does not remain constant but changes continuously.

Finally, a few comments are in order. The size of matrix $\mathbf{B}$ in Eq. (31) will be always 4×4 , regardless of the number of discontinuity points. Also, the eigenfunctions ψ , θ , μ and χ corresponding to each eigenvalue ω will inherently satisfy the required conditions at the discontinuity points ξ_j 's: that is, while deflection ψ and bending moment μ will be continuous over the whole domain, θ will be discontinuous at RJs' locations, and χ will be discontinuous at TSs' or LMs' locations.

4. Orthogonality conditions

In this section, orthogonality conditions will be derived for the deflection eigenfunctions ψ , generalizing the approach devised by Oliveto et al. (1997) for a EB beam with end viscous RJs. First, Eq. (12) is written for the m th and n th eigenfunctions ψ_m and ψ_n as

$$\frac{d^4 \psi_m}{d\xi^4} + \sum_{j=1}^N (\kappa_{eq,j}^{(u)}(\omega_m) - \Pi_j \omega_m^2) \psi_m(\xi_j) \delta(\xi - \xi_j) + \sum_{j=1}^N (\kappa_{eq,j}^{(\theta)}(\omega_m))^{-1} \mu_m(\xi_j) \delta^{(2)}(\xi - \xi_j) - \omega_m^2 \psi_m(\xi) = 0 \quad (36)$$

$$\frac{\bar{d}^4 \psi_n}{d\xi^4} + \sum_{j=1}^N (\kappa_{eq,j}^{(u)}(\omega_n) - \Pi_j \omega_n^2) \psi_n(\xi_j) \delta(\xi - \xi_j) + \sum_{j=1}^N (\kappa_{eq,j}^{(\theta)}(\omega_n))^{-1} \mu_n(\xi_j) \delta^{(2)}(\xi - \xi_j) - \omega_n^2 \psi_n(\xi) = 0 \quad (37)$$

Multiplying Eq. (36) by ψ_n and integrating over $[0,1]$ lead to

$$\int_0^1 \frac{\bar{d}^2 \psi_m}{d\xi^2} \frac{\bar{d}^2 \psi_n}{d\xi^2} d\xi + \left\{ \sum_{j=1}^N \kappa_{eq,j}^{(u)}(\omega_m) \psi_m(\xi_j) \psi_n(\xi_j) - (\kappa_{eq,j}^{(\theta)}(\omega_m))^{-1} \mu_m(\xi_j) \mu_n(\xi_j) + P_j(\omega_m, \omega_n) \right\} - \omega_m^2 \left(\int_0^1 \psi_m \psi_n d\xi + \sum_{j=1}^N \Pi_j \psi_m(\xi_j) \psi_n(\xi_j) \right) = 0, \quad \text{for} \quad (38)$$

$$P_j(\omega_m, \omega_n) = -(\kappa_{eq,j}^{(\theta)}(\omega_m))^{-1} (\kappa_{eq,j}^{(\theta)}(\omega_n))^{-1} \cdot \mu_m(\xi_j) \int_0^1 \mu_n(\xi_j) \delta(\xi - \xi_j) \delta(\xi - \xi_j) d\xi \quad (39)$$

Notice that Eq. (38) can readily be derived using integration by parts and the sampling property of Dirac's delta (Falsone, 2002). Similarly, multiplying Eq. (37) by ψ_m and integrating over $[0,1]$ lead to

$$\int_0^1 \frac{\bar{d}^2 \psi_m}{d\xi^2} \frac{\bar{d}^2 \psi_n}{d\xi^2} d\xi + \left\{ \sum_{j=1}^N \kappa_{eq,j}^{(u)}(\omega_n) \psi_m(\xi_j) \psi_n(\xi_j) - (\kappa_{eq,j}^{(\theta)}(\omega_n))^{-1} \mu_m(\xi_j) \mu_n(\xi_j) + P_j(\omega_m, \omega_n) \right\} - \omega_n^2 \left(\int_0^1 \psi_m \psi_n d\xi + \sum_{j=1}^N \Pi_j \psi_m(\xi_j) \psi_n(\xi_j) \right) = 0 \quad (40)$$

The difference between Eq. (40) and Eq. (38) yields the first orthogonality condition

$$(\omega_m^2 - \omega_n^2) \left(\int_0^1 \psi_m \psi_n d\xi + \sum_{j=1}^N \Pi_j \psi_m(\xi_j) \psi_n(\xi_j) \right) + O_1(\omega_m, \omega_n) = 0, \quad \text{for} \quad (41)$$

$$O_1(\omega_m, \omega_n) = \sum_{j=1}^N \{ [\kappa_{eq,j}^{(u)}(\omega_n) - \kappa_{eq,j}^{(u)}(\omega_m)] \psi_m(\xi_j) \psi_n(\xi_j) + [(\kappa_{eq,j}^{(u)}(\omega_m))^{-1} - (\kappa_{eq,j}^{(u)}(\omega_n))^{-1}] \cdot \mu_m(\xi_j) \mu_n(\xi_j) \} \quad (42)$$

Further, computing the difference between Eq. (40) as multiplied by ω_m and Eq. (38) as multiplied by ω_n leads to the second orthogonality condition

$$(\omega_m - \omega_n) \int_0^1 \frac{\bar{d}^2 \psi_m}{d\xi^2} \frac{\bar{d}^2 \psi_n}{d\xi^2} d\xi + \omega_m \omega_n (\omega_m - \omega_n) \left(\int_0^1 \psi_m \psi_n d\xi + \sum_{j=1}^N \Pi_j \psi_m(\xi_j) \psi_n(\xi_j) \right) + O_2(\omega_m, \omega_n) = 0, \quad \text{for} \quad (43)$$

$$O_2(\omega_m, \omega_n) = \sum_{j=1}^N \{ [\omega_m \kappa_{eq,j}^{(u)}(\omega_n) - \omega_n \kappa_{eq,j}^{(u)}(\omega_m)] \psi_m(\xi_j) \psi_n(\xi_j) + [\omega_n (\kappa_{eq,j}^{(\theta)}(\omega_m))^{-1} - \omega_m (\kappa_{eq,j}^{(\theta)}(\omega_n))^{-1}] \mu_m(\xi_j) \mu_n(\xi_j) + (\omega_m - \omega_n) P_j(\omega_m, \omega_n) \} \quad (44)$$

5. Forced vibrations

Next, the orthogonality conditions (41)–(43) will be used to build the response to an arbitrary loading, using the complex modal superposition principle as previously devised by Oliveto et al. (1997) for a EB beam with end viscous RJs. Consider the motion equation

$$\frac{\bar{\partial}^4 h}{\partial \xi^4} + \sum_{j=1}^N f_j(\xi, \tau) + \ddot{h}(\xi, \tau) = \varphi_\xi(\xi) \delta(\tau) \quad (45)$$

where $\varphi_\xi(\xi) \delta(\tau)$ is a space-dependent impulsive loading at $\tau = 0$, with $\varphi_\xi(\xi)$ being an arbitrary space-dependent function; $h(\xi, \tau)$ denotes the dimensionless IRF and $f_j(\xi, \tau)$ is given as

$$f_j(\xi, \tau) = [\kappa_j^{(u)} h(\xi_j, \tau) + \gamma_j^{(u)} \dot{h}(\xi_j, \tau) + \Pi_j \ddot{h}(\xi_j, \tau)] \delta(\xi - \xi_j) - \left(\frac{\bar{\partial} h}{\partial \xi} \Big|_{\xi=\xi_j^+} - \frac{\bar{\partial} h}{\partial \xi} \Big|_{\xi=\xi_j^-} \right) \delta^{(2)}(\xi - \xi_j) \quad (46)$$

According to the complex modal superposition principle, the sought IRF $h(\xi, \tau)$ can be represented as (Oliveto et al., 1997)

$$h(\xi, \tau) = \sum_{k=1}^{\infty} h_k(\xi, \tau) = \sum_{k=1}^{\infty} g_k(\tau) \psi_k(\xi) \quad (47)$$

where symbol $h_k(\xi, \tau)$ denotes the MIRF of the k th mode, $\psi_k(\xi)$ is the k th deflection function, and due to the impulsive nature of the loading $g_k(\tau)$ can be cast as

$$g_k(\tau) = \bar{g}_k e^{i\omega_k \tau} \quad (48)$$

being $\bar{g}_k$ constant amplitudes. Upon replacing Eq. (47) in Eq. (45), multiplying Eq. (45) by ψ_m , integrating over $[0, 1]$ and using the orthogonality condition (43) leads to

$$\sum_{k=1}^{\infty} \frac{\ddot{g}_k}{\omega_k^2} \cdot \sum_{j=1}^N \left\{ \omega_k \frac{\kappa_{eq,j}^{(u)}(\omega_k) - \kappa_{eq,j}^{(u)}(\omega_m)}{(\omega_m - \omega_k)} \psi_m(\xi_j) \psi_k(\xi_j) + \omega_k \frac{(\kappa_{eq,j}^{(\theta)}(\omega_m))^{-1} - (\kappa_{eq,j}^{(\theta)}(\omega_k))^{-1}}{\omega_m - \omega_k} \mu_m(\xi_j) \mu_k(\xi_j) \right. \\ \left. + (\omega_m \omega_k + \omega_k^2) \left(\int_0^1 \psi_m \psi_k d\xi + \sum_{j=1}^N \Pi_j \psi_m(\xi_j) \psi_k(\xi_j) \right) \right\} = \eta_m \delta(\tau) \quad (49)$$

where integration by parts has been performed, $\ddot{g}_k = -\omega_k^2 g_k$ has been set due to Eq. (48) and

$$\eta_m = \int_0^1 \psi_m(\xi) \varphi_{\xi}(\xi) d\xi \quad (50)$$

It can readily be seen that in light of the orthogonality condition (41) the expression in curly brackets is equal to zero if $\omega_m \neq \omega_k$. Consequently the following uncoupled equations are obtained in $g_k(\tau)$:

$$\ddot{g}_k \Xi_k = \eta_k \delta(\tau), \quad \text{for} \quad \Xi_k = 2 \int_0^1 \psi_k^2 d\xi + 2 \sum_{j=1}^N \Pi_j \psi_k^2(\xi_j) - \frac{i\gamma_j^{(\theta)}}{\omega_k} \left[\frac{\mu_k(\xi_j)}{\kappa_{eq,j}^{(\theta)}(\omega_k)} \right]^2 \quad (51a,b)$$

Integrating Eq. (51a) over the time domain yields $\dot{g}_k(0^+) = \Xi_k^{-1} \eta_k$. Based on this result, taking into account that $\dot{g}_k(\tau) = i\omega_k g_k(\tau)$, the following relation is obtained for the constant amplitudes $\bar{g}_k$:

$$\bar{g}_k = (i\omega_k \Xi_k)^{-1} \eta_k \quad (52)$$

Upon replacing Eq. (52) in Eq. (48), the IRF (47) is finally determined as superposition of the MIRFs $h_k(\xi, \tau)$. Hence, the dimensionless response $\hat{u}(\xi, \tau) = u(x, t)/L$ to an arbitrary loading $\varphi(\xi, \tau) = \varphi_{\xi}(\xi) \varphi_{\tau}(\tau)$ can be computed by the convolution integral

$$\hat{u}(\xi, \tau) = \sum_{k=1}^{\infty} \int_0^{\tau} h_k(\xi, \tau') \varphi_{\tau}(\tau') d\tau' \quad (53)$$

where $h_k(\xi, \tau)$ involves the space-dependent part $\varphi_{\xi}(\xi)$ of $\varphi(\xi, \tau)$, see Eq. (48), Eq. (52) and Eq. (50). In practical applications, a few n modes in Eq. (53) will be generally sufficient to accurately represent the response. In this context, the MIRFs $h_k(\xi, \tau)$ associated with complex-conjugate eigenfunction pairs can be summed up and reverted to the following real form (Oliveto et al., 1997):

$$h_k^r(\xi, \tau) = \alpha_k(\xi) |\omega_k| h_k(\xi, \tau) + \beta_k(\xi) \dot{h}_k(\xi, \tau), \quad \text{for} \quad (54)$$

$$\alpha_k(\xi) = \zeta_k \beta_k(\xi) - \sqrt{1 - \zeta_k^2} \lambda_k(\xi); \quad \beta_k(\xi) = 2\text{Re}[\bar{g}_k \psi_k(\xi)]; \quad \lambda_k(\xi) = 2\text{Im}[\bar{g}_k \psi_k(\xi)] \quad (55a,b,c)$$

$$h_k(\xi, \tau) = \frac{1}{\omega_{Dk}} e^{-\zeta_k |\omega_k| \tau} \sin(\omega_{Dk} \tau); \quad \omega_{Dk} = |\omega_k| \sqrt{1 - \zeta_k^2} \quad (56a,b)$$

In frequency domain, the following MFRF corresponds to each MIRF $h_k^r(\xi, \tau)$ (Oliveto et al., 1997):

$$H_k^r(\xi, \omega) = \alpha_k(\xi) |H_k(\omega) + \beta_k(\xi) \dot{H}_k(\omega)|, \quad \text{for} \quad (57)$$

$$H_k(\omega) = (|\omega_k|^2 - \omega^2 + i2\zeta_k |\omega_k| \omega)^{-1}; \quad \dot{H}_k(\omega) = i\omega (|\omega_k|^2 - \omega^2 + i2\zeta_k |\omega_k| \omega)^{-1} \quad (58a,b)$$

Based on the MFRFs (57), the FRF can be obtained as (Oliveto et al., 1997)

$$H(\xi, \omega) \approx \sum_{k=1}^n H_k^r(\xi, \omega) \quad (59)$$

Eq. (59) yields the steady-state response of the EB discontinuous beam, subjected to a unit complex harmonic loading $1 \times \exp(i\omega \tau)$ with space distribution $\varphi_{\xi}(\xi)$. The corresponding exact solution can be built by a classical approach, i.e. expressing the steady-state response in a trigonometric form over uniform segments between consecutive discontinuity points, and enforcing matching conditions at the discontinuity points + B.C. (Craig, 1981). However, the exact solution has to be re-built for any value of the forcing frequency ω , while Eq. (59) is an analytical solution available for any value of ξ and ω .

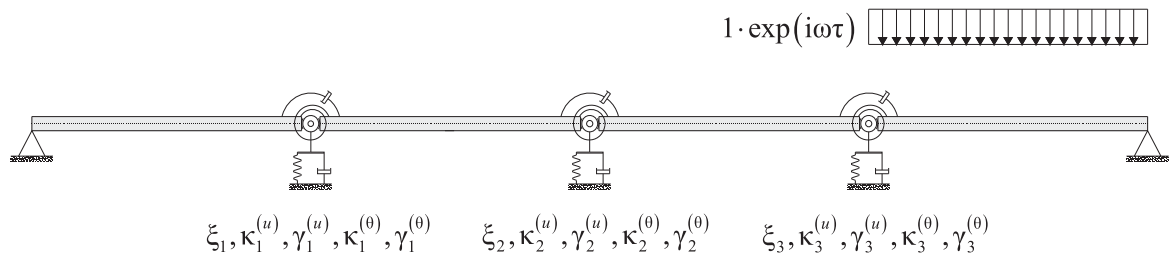


Fig. 2. Simply supported beam with KV viscoelastic TSs and RJs.

Table 1
Simply supported beam in Fig. 2: complex eigenvalues.

Mode	Eigenvalue	Damping ratio
1	$\pm 21.9037 + i0.2629$	0.0120
2	$\pm 39.8987 + i1.5487$	0.0388
3	$\pm 81.8244 + i6.5410$	0.0796
4	± 157.9140	0.0
5	$\pm 238.023 + i14.4789$	0.0607

6. Numerical applications

Two numerical examples are presented. The obtained eigenvalues and eigenfunctions have been compared with those built by the classical approach (see Section 1). Since the proposed method yields indeed the exact eigensolution, coincident results have been always found.

6.1. Simply supported beam with TSs and RJs

Consider the simply supported beam in Fig. 2. KV viscoelastic TSs and RJs occur at $\xi_1 = 0.25$, $\xi_2 = 0.50$ and $\xi_3 = 0.75$, modeling respectively viscoelastic external dampers and internal connections between the adjacent beam spans (Shukla and Datta, 1999; Sekulovic et al., 2002). The following parameters are taken: $\kappa_j^{(u)} = \kappa^{(u)} = 10^2$, $\gamma_j^{(u)} = \gamma^{(u)} = 10^{-1}$ for the TSs; $\kappa_j^{(\theta)} = \kappa^{(\theta)} = 10^1$, $\gamma_j^{(\theta)} = \gamma^{(\theta)} = 10^{-1}$ for the RJs.

Table 1 reports the dimensionless eigenvalues and the corresponding modal damping ratios of the first five modes, as computed by Netwon's method (Mathematica, 2008). All modes are complex, except for the 4th that is real. In fact the TSs and RJs occur at zero points in the deflection and bending moment of the bare beam 4th mode and, for this, neither the TSs nor the RJs are activated. It can also be seen that damping effects on the individual modes are not the same. The eigenfunctions of modes 1–2 are shown in Fig. 3, for some response variables (complex-conjugate eigenfunctions are not reported for conciseness): the rotation is discontinuous at the RJs (Fig. 3b) and the shear force is discontinuous at the TSs (Fig. 3d). Same considerations hold for the eigenfunctions of all other modes, here not reported for brevity.

Next, assume that a unit complex harmonic loading $1 \times \exp(i\omega\tau)$ is uniformly distributed over the 4th span, as shown in Fig. 2. The FRF (59) is computed by retaining the first $n=5$ modes, being $\varphi_\xi(\xi) = 0$ for $0 \leq \xi \leq 0.75$ and $\varphi_\xi(\xi) = 1$ for $0.75 \leq \xi \leq 1.0$. Fig. 4 shows the comparison with the exact classical solution (see end of Section 5). The agreement is very satisfactory for both the forcing frequencies $\omega = 20$ and $\omega = 40$, for which the response is dominated by mode 1 and 2, respectively (see Table 1). The proposed closed-form FRF (59) is more efficient than the exact classical solution that is a numerical solution to be re-built for each value of the forcing frequency ω .

The time-domain response to the 4th-span complex harmonic loading $1 \times \exp(i\omega\tau)$ in Fig. 2 can be computed using Eq. (53). In this case, the integrals in Eq. (53) can readily be solved in a closed form (Oliveto et al., 1997). Considering that $\exp(i\omega\tau) = \cos(\omega\tau) + i\sin(\omega\tau)$, the response features a real part and an imaginary part representing the responses to 4th-span harmonic loadings $\cos(\omega\tau)$ and $\sin(\omega\tau)$, respectively. Fig. 5 shows the real part of the response (=response to the 4th-span harmonic loading $\cos(\omega\tau)$) at location $\xi = 0.5$ (mid-span deflection), for $\omega = 20$ and $\omega = 40$, as obtained: (i) by Eq. (53) in a closed form, with the first $n = 5$ modes; (ii) by the finite element (FE) method, as implemented in ADINA (2013), with a mesh of 20 two-node elements and using a Newmark integration scheme. A very satisfactory agreement is found between Eq. (53) and the FE solution, for both $\omega = 20$ and $\omega = 40$. Accurate results can be found also for the imaginary part of the response at location $\xi = 0.5$, and at all other locations ξ . Such results reflect the accuracy of the FRF (59) over the whole beam axis (Fig. 4).

6.2. Cantilever beam with TSs and tip LM

Consider the cantilever beam studied by Gürgöze and Erol (2002), with a fixed support at $\xi_1 = 0.5$, a viscous damper at $\xi_2 = 0.75$ with dimensionless damping coefficient = 0.5 and a tip dimensionless mass = 1.0 at $\xi_3 = 1.0$. Applying the proposed method, a linearly elastic TS with $\kappa_1^{(u)} = 10^6$ at $\xi_1 = 0.5$, a viscous TS with $\gamma_2^{(u)} = 0.5$ at $\xi_2 = 0.75$ and a tip LM with $\Pi_3 = 1.0$ at $\xi_3 = 1.0^- = 0.999$ are considered.

Table 2 shows the dimensionless eigenvalues and damping ratios of the first five modes, as computed by Netwon's method (Mathematica, 2008). They are all complex, with different modal damping ratios. The FRF due to a unit complex harmonic loading $1 \times \exp(i\omega\tau)$ at the tip, as computed by retaining the first $n = 5$ modes in Eq. (59), is reported in Fig. 6 along with the exact classical solution (see end of Section 5). Again, the agreement is satisfactory for both the considered forcing frequencies, i.e. $\omega = 5$ and $\omega = 40$. Comments on the advantages of the FRF (59) with respect to the classical solution are not repeated for conciseness.

The time-domain response to the tip complex harmonic loading $1 \times \exp(i\omega\tau)$ can be computed by using Eq. (53), in a closed form (Oliveto et al., 1997). Figs. 7 and 8 show the imaginary part of the response (=response to tip harmonic loading $\sin(\omega\tau)$) at location $\xi = 1$

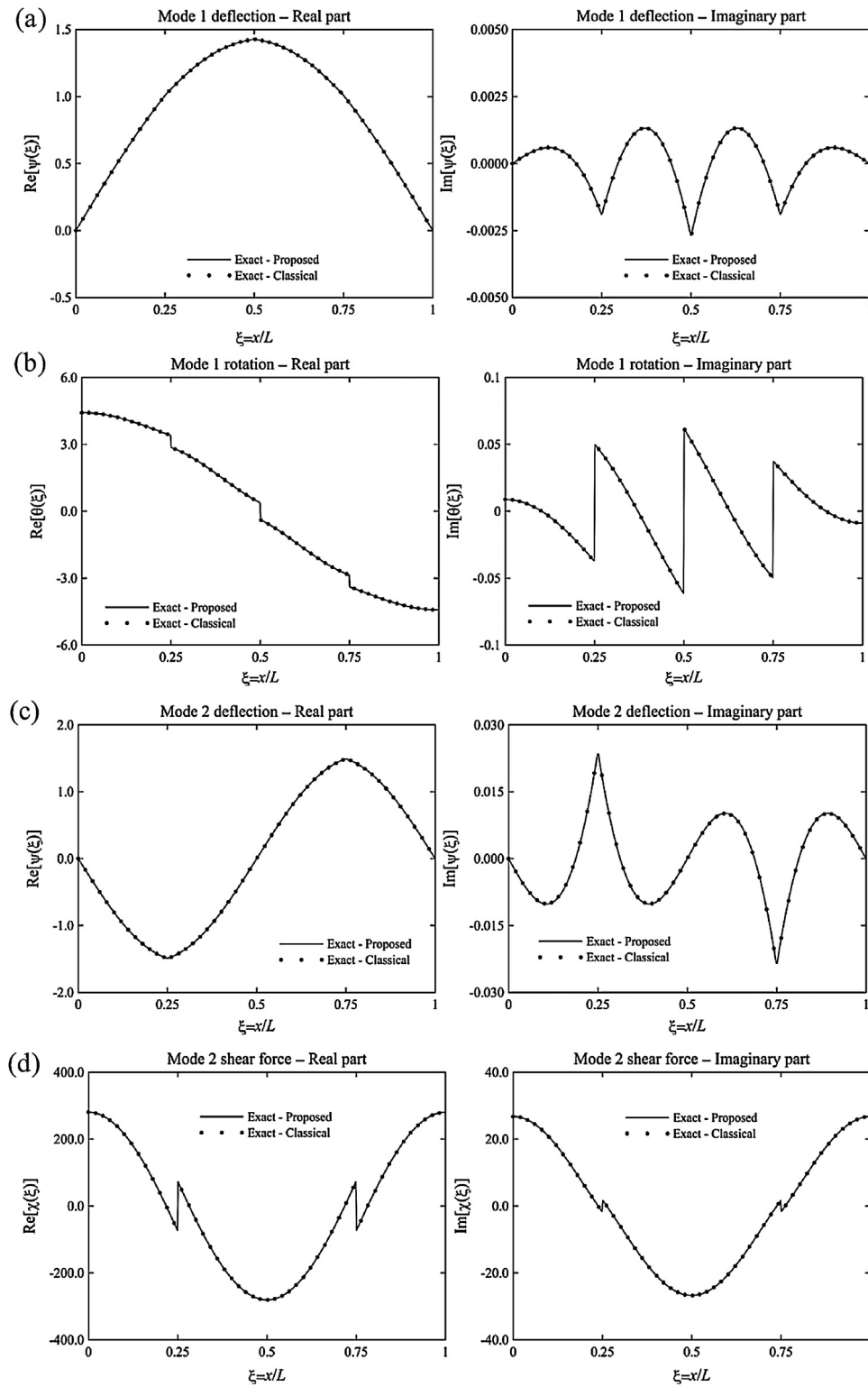


Fig. 3. Simply supported beam in Fig. 2: (a and b) deflection and rotation eigenfunctions of mode 1; (c and d) deflection and shear-force eigenfunctions of mode 2.

Table 2
Cantilever beam in paper by Gürgöze and Erol (2002): complex eigenvalues.

Mode	Eigenvalue	Damping ratio
1	$\pm 3.4794 + i0.0346$	0.0099
2	$\pm 47.6195 + i0.8197$	0.0172
3	$\pm 80.0977 + i0.1859$	0.0023
4	$\pm 173.3690 + i0.0095$	0.0001
5	$\pm 230.8020 + i0.0769$	0.0003

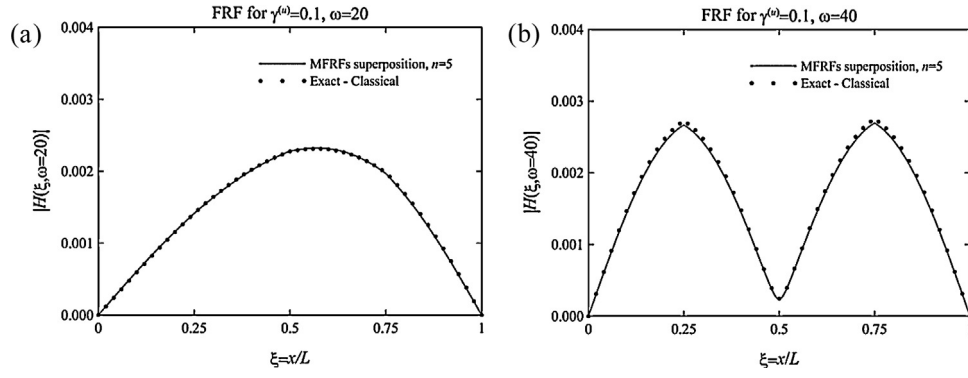


Fig. 4. Simply supported beam in Fig. 2 under a 4th span complex harmonic loading $1 \times \exp(i\omega\tau)$: FRF for (a) $\omega = 20$, (b) $\omega = 40$.

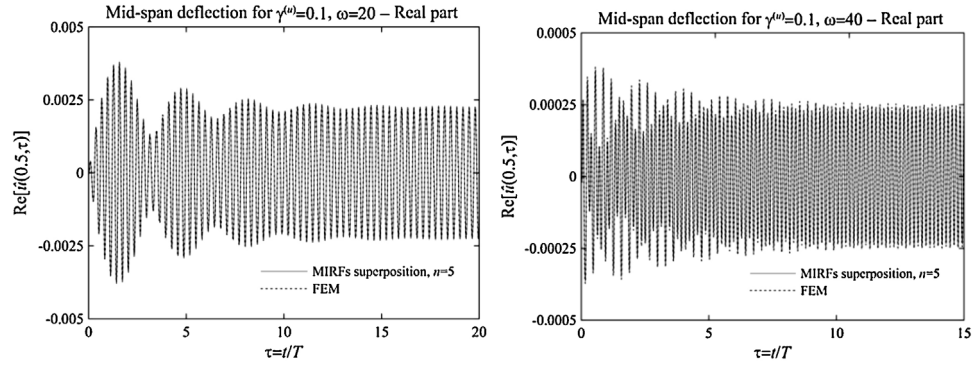


Fig. 5. Simply supported beam in Fig. 2: real part of the mid-span deflection under a 4th span complex harmonic loading $1 \times \exp(i\omega\tau)$ with (a) $\omega = 20$ and (b) $\omega = 40$.

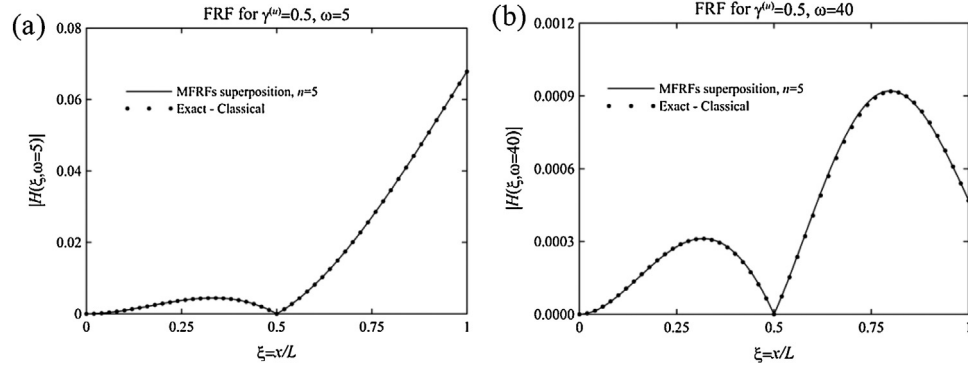


Fig. 6. Cantilever beam studied by Gürgöze and Erol (2002), under a tip complex harmonic loading $1 \times \exp(i\omega\tau)$: FRF with (a) $\omega = 5$ and (b) $\omega = 40$.

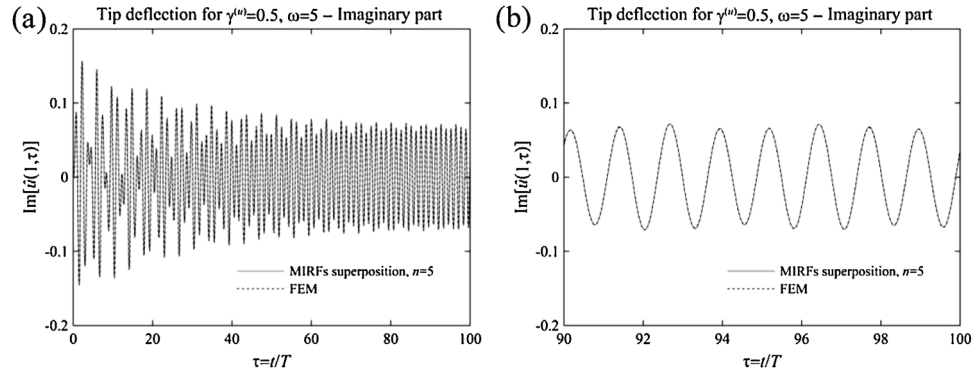


Fig. 7. Cantilever beam studied by Gürgöze and Erol (2002): imaginary part of the tip deflection under a tip complex harmonic loading $1 \times \exp(i\omega\tau)$ with $\omega = 5$, for (a) $0 < \tau < 100$ and (b) $90 < \tau < 100$.

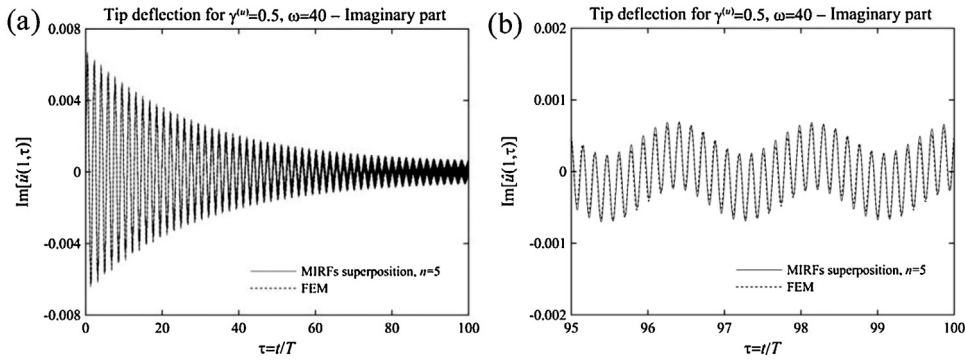


Fig. 8. Cantilever beam studied by Gürgöze and Erol (2002): imaginary part of the tip deflection under a tip complex harmonic loading $1 \times \exp(i\omega\tau)$ with $\omega = 40$, for (a) $0 < \tau < 100$ and (b) $95 < \tau < 100$.

(tip deflection), for the forcing frequencies $\omega = 5$ and $\omega = 40$ in Fig. 6, as obtained: (i) by Eq. (53) in a closed form; (ii) by the FE method implemented in ADINA (2013), with 20 two-node elements and using a Newmark integration scheme. The agreement is very satisfactory. Very accurate results can be found also for the real part of the response at location $\xi = 1$, and for all other locations ξ , as expected in view of the accuracy of the FRF (59) over the whole beam axis (Fig. 6).

7. Conclusions and further developments

The dynamics of beams with an arbitrary number of KV viscoelastic RJs, TSs and LMs has been addressed. The theory of generalized functions has been used to derive exact closed-form expressions of the eigenfunctions, inherently fulfilling the required conditions at the discontinuity points, and the exact CE as determinant of a 4×4 matrix, for any number of discontinuity points. Orthogonality conditions have been set for the eigenfunctions and used to build closed-form expressions of the IRF and FRF, for an efficient solution to forced vibrations by the complex modal superposition principle. The derivation of the orthogonality conditions, IRF and FRF, as well as the implementation of the complex modal superposition principle, can be considered as a generalization, to EB discontinuous beams with an arbitrary number of KV viscoelastic RJs, TSs and LMs, of the approach devised by Oliveto et al. (1997) for a EB beam with end viscous RJs.

The proposed solutions to forced vibrations involve some advantages. The FRF (59) is an analytical expression to be computed for any value of ω and ξ , while the classical solution is a numerical solution, to be re-built for any value of ω (see end of Section 5). The IRF (47) is also an analytical expression, leading to closed-form solutions of the convolution integrals (53) for harmonic functions (Oliveto et al., 1997). Considering that excitations generally involved in structural applications can be reverted to the sum of harmonic functions by Fourier transforming, such closed-form solutions are quite advantageous, for instance with respect to FE numerical solutions.

The proposed method can be extended to other types of discontinuities that can be all handled by the theory of generalized functions (Failla, 2011). For a comprehensive description of all possible discontinuities and pertinent solutions by the theory of generalized functions, the interested readers are referred to a recent paper by Caddemi et al. (2013b). A relevant further development is the derivation of the exact closed-form dynamic stiffness matrix of the beam. For this, the procedure is analogous to that proposed by Caddemi and Caliò (2013) for a beam with linearly elastic rotational springs. That is, a vector of nodal displacements/rotations $\mathbf{u}$ and a vector of nodal forces/moments $\mathbf{F}$ can be built as

$$\mathbf{u} = [\psi(0) \quad \theta(0) \quad \psi(1) \quad \theta(1)]^T = \mathbf{C}(\omega)\mathbf{a}, \quad \mathbf{F} = [-\chi(0) \quad \mu(0) \quad \chi(1) \quad -\mu(1)]^T = \mathbf{D}(\omega)\mathbf{a} \quad (60a,b)$$

Coefficients $c_{ij}(\omega)$'s and $d_{ij}(\omega)$'s in Eqs. (60) can be derived as closed-form functions of the eigenvalue ω in any symbolic package: $c_{ij}(\omega)$'s based on Eq. (17) for $\psi(\xi)$ and Eq. (19) for $\theta(\xi)$, and $d_{ij}(\omega)$'s based on Eq. (21) for $\mu(\xi)$ and Eq. (23) for $\chi(\xi)$, where v_j 's and ρ_j 's are computed by Eqs. (30). Hence, the exact dynamic stiffness matrix $\mathbf{K}(\omega)$ can be obtained as

$$\mathbf{F} = \mathbf{D}(\omega)\mathbf{a} = \mathbf{D}(\omega)\mathbf{C}^{-1}(\omega)\mathbf{u} = \mathbf{K}(\omega)\mathbf{u} \quad (61)$$

The 4×4 matrix $\mathbf{C}(\omega)$ can be inverted in a symbolic form, see example in the paper by Caddemi and Caliò (2013). $\mathbf{K}(\omega)$ could be used for an exact modal analysis of frame structures where each frame element between beam-column nodes could carry an arbitrary number of RJs, TSs and LMs, with significant computational advantages with respect to a standard FE approach, where a frame element should be considered between two consecutive discontinuity points. Such a computational advantage has been shown by Caddemi et al. (2013c), who have recently proposed a novel Timoshenko beam element with discontinuities for the analysis of discontinuous frames.

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